

Research Statement

Jai Moondra

How do we design good optimization algorithms for under-specified problems?

Under-specified objective functions arise in policy problems because stakeholders often disagree on the objective, and there is no ‘right’ way of balancing their priorities. This is the central question my Ph.D. thesis [P1] formalized: can algorithms give decision-makers a small set or *portfolio* of options to help them make the eventual choice, but without losing robust mathematical guarantees? The meaning of ‘good’ is subjective in such problems, and my research explores this in diverse areas: Is the response of an LLM ‘good’ when it prioritizes creativity or when it prioritizes accuracy [P13, P11]? Should algorithms that recommend locations for new facilities such as pharmacies and childcare centers prioritize the average person’s distance to facilities rather than the distances of those who are furthest away [P2, P3]? More generally, how do we deal with competing interests of stakeholders in resource allocation problems (e.g, in combinatorial problems such as scheduling [P4] and in reinforcement learning [P10])? Should algorithms prioritize speed (runtime), or solution quality (approximation guarantee), or robustness under uncertainty, or the ability to learn from historical data [P9, P5]? Such decisions are supposed to be made by the end user: the LLM user should get to decide what responses they prefer, and an executive or lawmaker should have the final say in how resources are allocated. And yet, automated systems must deal with these questions to produce any recommendation. Portfolios are motivated by this practical concern, but also lead to interesting mathematical and algorithmic problems.

Under-specified problems also arise when we seek to understand and improve emerging technologies, such as LLMs and quantum computing. For many problems, even formulating the problem mathematically is critical: a good formulation must make simplifying assumptions to be mathematically feasible without losing the core problem complexity. My research in quantum computing studies different aspects of this question, including issues of quantum noise, which must be dealt with before benefitting from the promised ‘quantum advantage’ over classical computers. Concretely, we illustrate using the Max-Cut problem that techniques from classical computing are greatly helpful in improving quantum algorithms [P6, P8, P7]. Alongside improvements in quantum hardware, this has helped mitigate noise and scale current noisy intermediate-scale quantum (NISQ) devices to larger instances for Max-Cut and related problems. A similar phenomenon is occurring in the very different world of machine learning. Here, my research shows that combinatorial and algorithmic ideas can significantly improve learning algorithms, including for optimization [P5], recommendation systems [P9], and personalization [P13, P10, P11]. In both cases, better theoretical understanding of the problem structure is helping to optimize the gains made by newer tools.

I aim to design theoretically robust algorithms that both use and aid emerging technologies. Next, I describe my research in more detail, broadly categorized into three areas: discrete optimization (§1), machine learning (§2), and quantum computing (§3).

1. DISCRETE OPTIMIZATION

The idea of *portfolios* is simple: many real-world optimization problems have a well-defined constraint set, but the objective is under-specified, often because it needs to balance competing objectives. We introduced the notion of a portfolio [P2], which is a small collection of feasible solutions such that irrespective of how the stakeholder objectives are balanced, one of these solutions is guaranteed to be approximately optimal for the resulting objective. Formally, given a class $\mathbf{C}$ of objective functions (representing different ways to balance competing stakeholder objectives), a collection P of solutions is called an α -approximate portfolio if for all objectives f in this class, there is some solution in P that is α -approximately optimal for f . This generalizes the well-studied notion of simultaneous approximations, but gives greater flexibility when no single solution is good for all objectives. As we

show in [P2, P4, P10], allowing a small number of solutions drastically improves the approximation guarantee for large – and even infinite – classes of objectives. This trade-off between the size of the portfolio and the approximation factor can be studied for any class of objectives for any optimization problem and constraint set.

In particular, consider problems such as facility location and scheduling, where each feasible solution induces a vector of different costs for each individual: distance to the nearest facility in facility location, and loads on different machines in scheduling. Minimizing the total cost is a very different problem than minimizing the maximum cost. Using other ways of balancing the cost vector leads to other problems, and a large body of literature has considered minimizing different norms of the cost vector, where each norm corresponds to a different notion of ‘balance’. In [P4], we consider the general class of *symmetric monotonic norms*, which includes any norm that is symmetric (invariant to swapping any two coordinates in the cost vector) and monotone (i.e., increasing any individual’s cost cannot decrease the norm). Most minimization problems in combinatorial optimization have a symmetric monotone norm as their objective, including variants of clustering (k -median, k -means, k -center), makespan minimization in scheduling, the traveling salesman and repairman problems, etc. We also study *ordered norms*, where each ordered norm is specified by some given weights, with the highest cost weighted by the highest weight, the second-highest cost weighted by the second-highest weight, and so on. Even with appropriate discretization, the number of ordered norms and symmetric monotonic norms increases exponentially with the dimension. It is therefore unclear if we can find a small portfolio.

For scheduling with identical jobs, we characterize the trade-off between portfolio size and approximation quality. More generally, we develop a new *primal-dual counting* framework: we show that the problem of computing portfolios for ordered norms is equivalent to counting in a suitably defined ‘dual space’. When this dual space is low-dimensional – such as for many scheduling problems – we get portfolios that work for exponentially many ordered norms but with only a polylogarithmic number of solutions. We also develop portfolios for a variety of other combinatorial problems.

Portfolios are also useful in practice: in [P2], we apply the machinery of portfolios to suggest ways to combat ‘medical deserts’: poor areas with limited or no access to medical facilities such as pharmacies. We document medical deserts in the U.S., and generate a portfolio of 3–5 plans to open new pharmacies to combat medical deserts. Each plan recommends 10 new pharmacies, and different plans optimize different objectives: some plans prioritize the most vulnerable populations, while others prioritize the average person. Despite this, each plan potentially reduces dozens of medical deserts, covering hundreds of thousands of people. This gives policymakers a clear set of good options that make the trade-offs clear. Our solutions are freely available to use through an online tool [P3].

2. MACHINE LEARNING AND ONLINE OPTIMIZATION

Many machine learning problems present similar challenges with competing objectives. For example, different users prefer different kinds of LLMs, leading to personalization and alignment problems. Many dynamic resource allocation problems are solved through reinforcement learning, and the RL algorithm’s objective encodes which group of people benefits most. Risk-assessment algorithms (such as credit approvals or criminal re-offense predictors) need to balance accuracy with fairness.

With my collaborators at MIT Sloan and Harvard University, I worked on portfolios for reinforcement learning algorithms for resource allocation [P10]. In particular, we consider the class of p -means objective functions, which generalize commonly used objectives such as utilitarian welfare (maximize resources for the average individual or group), egalitarian welfare (maximize resources for the neediest), and Nash social welfare. While RL algorithms apply to any given objective, the choice of the objective matters in practice, as we show. As in the purely combinatorial settings above, we

show how to construct portfolios, and establish a trade-off between the number of solutions in the portfolio and the portfolio’s approximation guarantee. However, this learning setting presents a new challenge not present in purely combinatorial settings: each call to an RL algorithm is expensive and time-consuming. Therefore, our portfolio algorithms also balance the number of such ‘oracle calls’ alongside portfolio size and approximation guarantee. We propose two algorithms with theoretical guarantees. Our algorithms produce diverse and high-quality solutions across three very different experimental settings: (1) resource allocation for an NGO that focuses on the education of young mothers in India, (2) simulations for a driverless taxi that automatically decides which customers to pick up, and (3) a simulated model for resource allocation after a natural disaster.

Other applications across learning problems include:

- (1) LLM personalization using weighted preferences [P11], where we give a portfolio algorithm to personalize LLM responses to each user’s preferences without training or fine-tuning a new LLM for every user.
- (2) Personalizing chatbot rankings [P13], where we argue that global chatbot rankings poorly represent the preferences of a significant number of users, and are particularly bad for many non-English-speaking cohorts. Instead, we construct 3 to 4 distinct rankings of chatbots, and show that these represent user preferences much better.
- (3) Online convex optimization [P5], where we adapt mirror descent to the problem instance using a portfolio of mirror maps based on block norms. In particular, we consider sparse losses which commonly arise in machine learning and combinatorial problems. We show that using block norm mirror maps for sparse losses leads to large improvements in regret guarantees for a variety of common constraint sets.
- (4) Learning over submodular polytopes [P9], where we improve runtimes for mirror descent variants by developing techniques to reuse repeated information. Using structural results for submodularity, we develop combinatorial tools that exploit information patterns. In particular, we model recommendation systems through learning over a specific submodular polytope – the permutahedron – and show that our tools speed up the learning algorithm by 2 to 6 orders of magnitude in different cases.

3. QUANTUM COMPUTING

For several problems, quantum algorithms are known to use fewer operations asymptotically than classical algorithms. However, quantifying the advantage of such algorithms has been open except for a handful of problems, and remains crucial to identifying opportunities for use of quantum computers. Further, even if these quantum algorithms are better in theory, the current noisy intermediate-scale quantum (NISQ) hardware is noisy, and presents additional challenges. This also makes the problem under-specified: there is no ‘correct’ noise model for given hardware, and the problem formulation critically determines how well we can handle noise on actual devices.

With my collaborators at Georgia Tech, Georgia Tech Research Institute (GTRI), and Oak Ridge National Laboratory (ORNL), I studied these issues in discrete optimization generally, and in particular for the celebrated Quantum Approximate Optimization Algorithm (QAOA) [R1] for Max-Cut and other related problems. QAOA and its variants are iterative ‘hybrid’ quantum-classical algorithms that converge to the optimal Max-Cut over an infinite horizon. However, its convergence rate is unknown, and its performance on real hardware remains a challenge.

A first issue is embedding the underlying graph into the quantum hardware [P6]. QAOA requires the graph to be input in a specific form, with the qubits entangled according to the graph’s *cost Hamiltonian* matrix, which encodes the graph’s edges and their weights. Unlike classical hardware

where the graph can be embedded using traditional data structures such as an adjacency list or matrix, very limited kinds of operations are allowed for most quantum hardware. Further, since quantum noise is a real concern, it is critical to use as few operations as possible. For trapped-ion hardware used at Georgia Tech, we translated the physical description of the problem into a graph-theoretic one. This led to an asymptotically faster algorithm (called Union-of-Stars) for embedding graphs, and it is used in practice on the trapped-ion hardware at Georgia Tech and ORNL. The previous naive construction embedded one edge at a time, and our faster construction helped scale QAOA to larger graphs. Our construction sparked a separate master’s thesis by S. Dinpazhouh at Rice University, and it remains an active research area. In a follow-up work [P8], we showed that we can speed up the construction even further by classical pre-processing of the graph that adapts it better for the embedding process, while preserving most of the graph’s cut structure. Our pre-processing uses classical graph-theoretic ideas – decomposing weighted graphs into unweighted parts and cut sparsification – and modifies them suitably to provide faster algorithms. For both simulations and constructions on actual quantum computers, our algorithms lead to a significant reduction of noise, showing that a better understanding of the combinatorial structure of the problem leads to tangible outcomes.

In [P7], we bring other classical techniques to QAOA and provide a partial answer to its convergence rate. We show that using the classical Max-Cut algorithm [R2] of Goemans and Williamson to ‘warm-start’ QAOA leads to non-trivial approximation ratios, the first such guarantees for arbitrary graphs. Further, this version of QAOA actually outperforms the GW algorithm for a certain family of graphs, where GW is known to be asymptotically tight [R3]. Our numerical experiments show that warm-started QAOA can produce higher-quality cuts than both standard QAOA and Goemans-Williamson on the tested instances, showing the benefits of combining classical and quantum techniques.

4. FUTURE DIRECTIONS

I am interested in pursuing research problems in both theoretical and applied research, and particularly those that are both mathematically interesting and impactful in practice.

One general direction I am interested in is developing good theoretical algorithms that can exploit the predictive power of modern machine learning. Better ML algorithms have made predictions cheaper, and using these predictions well becomes the bottleneck in many problems. One useful theoretical framework for this is *stochastic probing*, where we seek to probe unknown outcomes for random events or random variables, given only their distributions (predicted, for example, by an ML algorithm based on historical data). The goal is to find a suitable solution, while minimizing the probing cost. This is a useful paradigm for many practical problems. For example, in hiring problems, estimates for candidate quality are readily available through their resumes and recommendations, but knowing them better often requires meeting and interviewing them (probing). Retail firms use expensive inspections (probing) to decide where a new warehouse or retail store must be opened, but often have a rough idea of how expensive each site is. In recent work [P12], we develop stochastic probing algorithms for matroid-based objectives. With R. Ravi (CMU) and Viswanath Nagarajan (University of Michigan), I am exploring more flexible models for stochastic probing that allow for multiple predictions through different ML models.

Developing a better theoretical understanding of portfolios in different settings remains open, and I am studying portfolios in online combinatorial optimization with R. Ravi and Karan Singh as a postdoctoral fellow at CMU. Our work aims to unify several disparate problems across combinatorial optimization and online learning, with potential applications to online recommendation systems.

PUBLICATIONS

- [P1] **Jai Moondra**. New directions in multi-objective optimization with applications.
Ph.D. Thesis, *Georgia Institute of Technology*, December 2025.
- [P2] Swati Gupta, **Jai Moondra**, and Mohit Singh. Which L_p norm is the fairest? Approximations for fair facility location across all “ p ”.
In *Economics and Computation (EC)*, July 2023.
Under submission to journal.
- [P3] Swati Gupta, **Jai Moondra**, and Mohit Singh. Medical Deserts in USA, <https://usa-medical-deserts.streamlit.app/>
Web Tool, July 2023.
- [P4] Swati Gupta, **Jai Moondra**, and Mohit Singh. Balancing Notions of Equity: Trade-offs Between Fair Portfolio Sizes and Achievable Guarantees.
Symposium on Discrete Algorithms (SODA), January 2025.
Mathematical Programming A, June 2026.
- [P5] Swati Gupta, **Jai Moondra**, and Mohit Singh. Improved regret guarantees for online mirror descent using a portfolio of mirror maps.
Under submission to journal, August 2026.
- [P6] Joel Rajakumar, **Jai Moondra**, Bryan Gard, Swati Gupta, and Creston D. Herold. Generating target graph couplings for the quantum approximate optimization algorithm from native quantum hardware couplings.
Physical Review A, August 2022.
- [P7] Reuben Tate, **Jai Moondra**, Bryan Gard, Greg Mohler, and Swati Gupta. Warm-Started QAOA with Custom Mixers Provably Converges and Computationally Beats Goemans-Williamson’s Max-Cut at Low Circuit Depths.
Quantum, September 2023.
- [P8] **Jai Moondra**, Phillip C. Lotshaw, Greg Mohler, and Swati Gupta. Promise of graph sparsification and decomposition for noise reduction in QAOA: Analysis for trapped-ion compilations.
Quantum, July 2026.
- [P9] **Jai Moondra**, Hassan Mortagy, and Swati Gupta. Reusing Combinatorial Structure: Faster Iterative Projections over Submodular Base Polytopes.
Advances in Neural Information Processing Systems (NeurIPS), December 2021.
- [P10] Cheol Woo Kim, **Jai Moondra**, Shresth Verma, Madeleine Pollack, Lingkai Kong, Milind Tambe, and Swati Gupta. Navigating the Social Welfare Frontier: Portfolios for Multi-objective Reinforcement Learning.
In *International Conference on Machine Learning (ICML)*, July 2025.
- [P11] Cheol Woo Kim, **Jai Moondra**, Roozbeh Nahavandi, Andrew Perrault, Milind Tambe, and Swati Gupta. Many preferences, few policies: Towards scalable language model personalization.
Under submission to conference, January 2026.
- [P12] Rohan Ghuge, **Jai Moondra**, and Mohit Singh. Stochastic function certification with correlations.
Symposium on Discrete Algorithms (SODA), January 2027 (to appear).

- [P13] **Jai Moondra**, Ayela Chughtai, Bhargavi Lanka, and Swati Gupta. Why global LLM leaderboards are misleading: Small portfolios for heterogeneous supervised ML. *Under preparation*, May 2026.

REFERENCES

- [R1] Edward Farhi, Jeffrey Goldstone, and Sam Gutmann. A Quantum Approximate Optimization Algorithm, November 2014. arXiv:1411.4028.
- [R2] Michel Goemans and David P. Williamson. Improved approximation algorithms for maximum cut and satisfiability problems using semidefinite programming. *Journal of the ACM (JACM)*, November 1995.
- [R3] Howard Karloff. How Good is the Goemans-Williamson MAX CUT Algorithm? *SIAM Journal on Computing*, January 1999.